

七. QED

1. 旋量QED的Feynmann规则

Maxwell场传播子: $Z[J] = \int DA_\mu e^{i \int d^4x (L + J^\mu A_\mu)}$

$$\begin{aligned} F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu \\ L &= -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{1}{2} (\partial_\nu A_\mu) F^{\mu\nu} \stackrel{\text{全微分为0}}{=} \frac{1}{2} \partial_\nu (A_\mu F^{\mu\nu}) - \frac{1}{2} A_\mu \partial_\nu F^{\mu\nu} \\ &= -\frac{1}{2} A_\mu \partial_\nu F^{\mu\nu} = -\frac{1}{2} A_\mu \partial_\nu (\partial^\mu A^\nu - \partial^\nu A^\mu) = \frac{1}{2} A^\mu (\underbrace{g_{\mu\nu} \square - \partial_\mu \partial_\nu}_{K_{\mu\nu}}) A^\nu \\ Z[J] &= \int DA_\mu e^{i \int d^4x (\frac{1}{2} A^\mu K_{\mu\nu} A^\nu + J_\mu A^\mu)} = e^{-\frac{i}{2} J^\mu D_{\mu\nu} J^\nu} \end{aligned}$$

$K_{\mu\nu}$ —— 规范不变性: $K_{\mu\nu} \partial^\mu \chi = 0$

本征值为0 $\rightarrow K_{\mu\nu}^{-1}$ 不存在

$A^\mu + \partial^\mu \chi \rightarrow G(x,y) = \langle 0 | T A_\mu(x) A_\nu(y) | 0 \rangle = i D_{\mu\nu}(x-y)$
可取无穷

防发散: 取规范

对Lorentz-规范: $\partial_\mu A^\mu = 0 \rightarrow L = \frac{1}{2} A^\mu \underbrace{g_{\mu\nu} \square}_{K_{\mu\nu}} A^\nu \rightarrow D_{\mu\nu} = -g_{\mu\nu} \Delta_{F0}(x-y), \square \Delta_{F0}(x-y) = -\delta(x-y)$

$D_{\mu\nu}$: Feynmann传播子

展开: $\Delta_{F0}(x-y) = \int \frac{d^4k}{(2\pi)^4} \tilde{\Delta}_{F0}(k) e^{-ikx}$
 $\tilde{\Delta}_{F0} = \frac{1}{k^2 + i\epsilon}$
 $m=0 \text{ 的 } \Delta_F(x-y)$

规范固定项:

规范固定项

$$L = \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{\lambda}{2} (\partial_\mu A^\mu)^2 = L_0 + L_{GF}$$

$$= -\frac{1}{2} A_\mu \partial_\nu F^{\mu\nu} + \frac{\lambda}{2} A^\mu \partial_\mu \partial_\nu A^\nu = \frac{1}{2} A_\mu (\underbrace{g_{\mu\nu} \square}_{K_{\mu\nu}} + (\lambda-1) \partial_\mu \partial_\nu) A^\nu$$

$$(g^{\mu\rho} \square + (\lambda-1) \partial^\mu \partial^\rho) D_{F\rho\nu}(x-y) = g^\mu_\nu \delta(x-y), \text{ 展开 } D_{F\mu\nu}(x) = \int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2 + i\epsilon} \cdot [g_{\mu\nu} - (1-\lambda) \frac{k_\mu k_\nu}{k^2 + i\epsilon}] e^{-ikx}$$

$\lambda=1$: Feynmann规范, $\lambda=\infty$: Landau规范

Feynmann规则:

旋量场拉氏密度: $L = L_N + L_\gamma + L_I$

$$L_\gamma = \frac{1}{2} A^\mu K_{\mu\nu} A^\nu$$

L_I

要求定域规范不变: $L_N + L_I = \bar{\psi} (i \gamma^\mu D_\mu - m) \psi = \bar{\psi} (i \gamma^\mu \partial_\mu - m) \psi - \underbrace{q \bar{\psi} \gamma^\mu \psi A_\mu}_{L_I}$

$\downarrow$

生成泛函 $Z_0[J, \bar{\eta}, \eta] = Z_0[J] Z_0[\bar{\eta}, \eta] = e^{\frac{i}{2} \int d^4x J^2 - i \int d^4x \bar{\eta} \eta}$

$\downarrow$
 $Z[J, \bar{\eta}, \eta] = N e^{-i \int d^4x \bar{\eta} \gamma^\mu \frac{1}{i \not{\partial} - m} \eta} Z_0$

$\downarrow$
 S矩阵: $S = N \{ e^{-i \int d^4x \bar{\eta} \gamma^\mu \frac{1}{i \not{\partial} - m} Z[A_{\mu in}, \psi_{in}, \bar{\psi}_{in}; J, \bar{\eta}, \eta]} \}$

$Z[A_{\mu in}, \psi_{in}, \bar{\psi}_{in}; J, \bar{\eta}, \eta] = e^{i A_{\mu in} J^\mu - \frac{i}{2} J^\mu D_{\mu\nu} J^\nu} e^{i \bar{\psi}_{in} \eta + i \bar{\eta} \psi_{in} - i \bar{\eta} S \eta}$

$S_1 = -iq N \left[\frac{1}{i \not{\partial} - m} \gamma^\mu \frac{1}{i \not{\partial} - m} Z \right] = -iq \int d^4x \bar{\psi}_{in} \gamma^\mu \psi_{in} A_\mu =$ 电磁相互作用顶点

Feynmann图: 同标量场

! 待补到附录 动量空间: 外线: 流入 $k \ e_{k\mu}^\delta$
 内线: 规范动量 $\frac{-i}{k^2 + i\epsilon} \left[g_{\mu\nu} - (1-\lambda) \frac{k_\mu k_\nu}{k^2 + i\epsilon} \right]$ (离壳)
 电磁顶点: 积分图 $-iq(2\pi)^4 \delta^4(\Sigma p_i) \gamma^\mu$

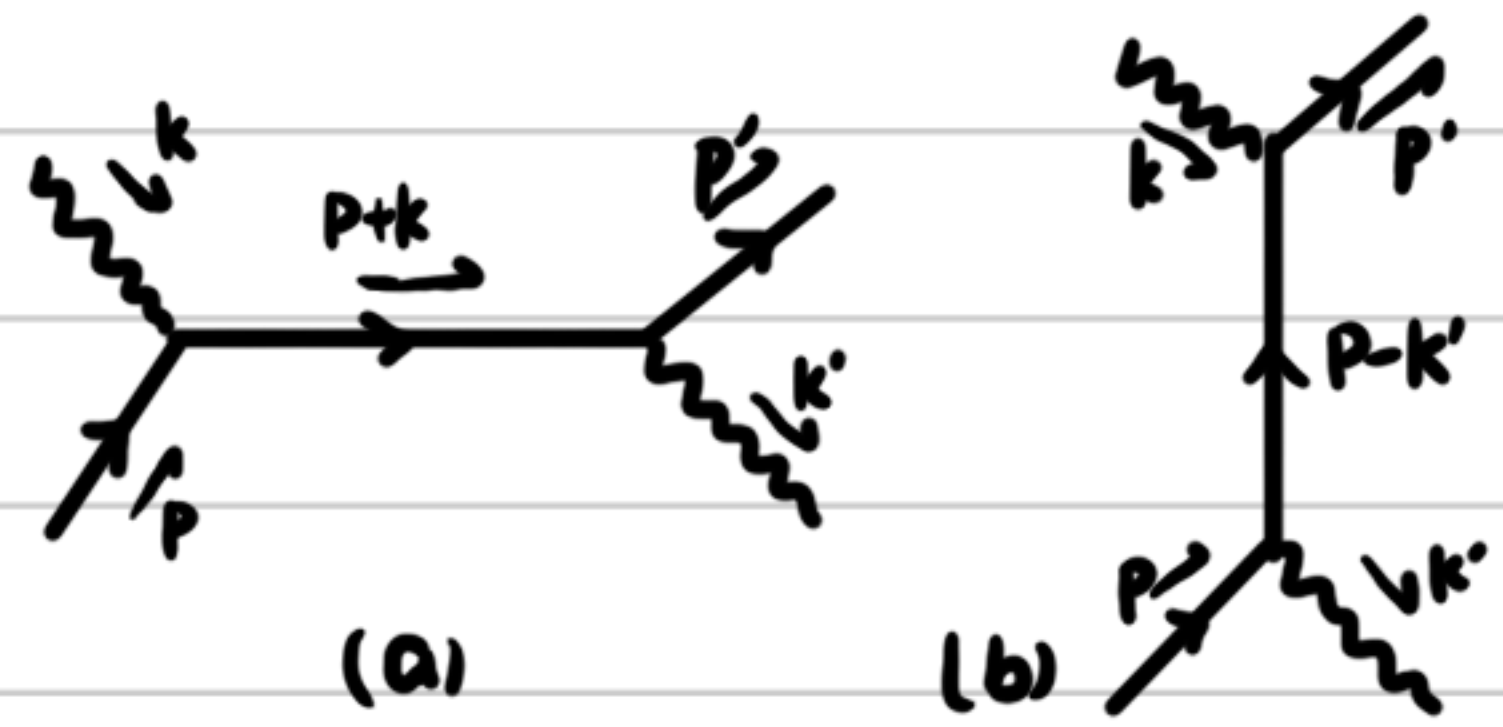
2. 几个QED过程:

1. Compton 散射: 物理过程: $\gamma + e^- \rightarrow \gamma' + e^-$ 能动关系: $k + 0 = k' + p'$
 $m^2 + 2m(\omega - \omega') - 2\omega\omega'(1 - \cos\theta) = m^2$

散射截面: $d\sigma = \frac{V d^3 p_{fi}}{V}$, $\bar{P}_{fi} = \frac{1}{4} \sum_{\text{spin}} \int \frac{V d^3 k'}{(2\pi)^3} \frac{V d^3 p'}{(2\pi)^3} V(2\pi)^4 \delta^4(p - p_i) |M_{fi}|^2$
 自旋与偏振平均

$E_f = \omega + p'_0$ 为总能
 $d^3 k = k^2 dk d\Omega = \frac{k^2 dk}{dE_f} dE_f d\Omega = \pi^2 p(E_f) dE_f d\Omega$
 $P(E_f) = \frac{2 \cdot 4\pi k^2 dk}{(2\pi)^3 dE_f} = \frac{\omega' d\omega'}{\pi^2 dE_f} \rightarrow$ 态密度 (E_f: 末态总能)
 $\frac{dE_f}{d\omega'} = 1 + \frac{dp'}{d\omega'} = \frac{m\omega}{p'_0 \omega'}$
 $P(E_f) = \frac{\omega'^3 p'_0}{\pi^2 m \omega}$

$\frac{d\sigma}{d\Omega} = \frac{V^4}{16V} P(E_f) \sum_{\text{spin}} |M_{fi}|^2$
 $\downarrow$
 $= \frac{1}{16\pi^2} \frac{\omega'^2}{m^2 \omega} \sum_{\text{spin}} |M_{fi}|^2$
 光子归一化 $N_{fi} = \frac{1}{4(p'_0 \omega' m \omega) V^2}$ ($v=1$)



求不变振幅:

展开到 S_2 : 2个图: $S_{fi(a)} = -q^2 (2\pi)^4 \delta(P_f - P_i) N_{fi} e_{k\mu}^{\delta'} \bar{u}(P', s') \gamma^\mu \frac{1}{\not{p} + \not{k} - m + i\epsilon} \gamma^\mu e_{k\mu}^{\delta} u(P, s)$

$S_{fi(b)} = -q^2 (2\pi)^4 \delta(P_f - P_i) N_{fi} e_{k\mu}^{\delta'} \bar{u}(P', s') \gamma^\mu \frac{1}{\not{p} - \not{k}' - m + i\epsilon} \gamma^\mu e_{k\mu}^{\delta} u(P, s)$

$M_{fi} = -iq^2 e_{k\mu}^{\delta'} \bar{u}(P', s') O^{\mu\nu} e_{k\mu}^{\delta} u(P, s)$

对称振幅求和:

$O^{\mu\nu} = \gamma^\mu \frac{1}{\not{p} + \not{k} - m + i\epsilon} \gamma^\nu + \gamma^\nu \frac{1}{\not{p} - \not{k}' - m + i\epsilon} \gamma^\mu$

$$\begin{aligned} \sum |M_{fi}|^2 &= q^4 |e_{k\mu}^{\delta'} \bar{u}(P', s') O^{\mu\nu} e_{k\mu}^{\delta} u(P, s)|^2 \\ &= q^4 \sum_{\delta\delta'} e_{k\mu}^{\delta'} e_{k\mu}^{\delta} \bar{u} O^{\mu\nu} u e_{k\nu}^{\delta'} e_{k\nu}^{\delta} \bar{u} O^{\nu\mu} u \\ &= q^4 \sum_{\delta\delta'} e_{k\mu}^{\delta'} e_{k\mu}^{\delta} e_{k\nu}^{\delta'} e_{k\nu}^{\delta} \cdot \sum_{ss'} [\bar{u}(P', s') O_{\mu\nu} u(P, s)]^\dagger [\bar{u}(P', s') O^{\nu\mu} u(P, s)] \\ &= q^4 g_{\mu\nu}^{\delta'} g_{\mu\nu}^{\delta} \sum_{ss'} \dots = q^4 \sum_{ss'} [\bar{u}(P', s') O_{\mu\nu} u(P, s)]^\dagger [\bar{u}(P', s') O^{\mu\nu} u(P, s)] \end{aligned}$$

标量光子与纵光子对散射 $\sum_\delta e_{k\delta}^{\delta} e_{k\delta}^{\delta} = g^{\mu\nu}$

振幅中与P抵消的项就相抵消

Lorentz不变: $L^\mu e_{k\mu}^{\delta}$, 变换 $e_{k\mu}^{\delta} \rightarrow k_\mu$, $L^\mu k_\mu = 0$

选 $k^\mu = (k^0, 0, 0, \omega)$, $k' = k^0 = \omega \rightarrow L^0 - L^3 = 0$

则 $\sum_\delta L^\mu e_{k\mu}^{\delta} = L^0 - L^3 = 0$

$\sum |M_{fi}|^2 = q^4 \sum_{ss'} [\bar{u} O_{\mu\nu} u]^\dagger [\bar{u} O^{\mu\nu} u]$

$\begin{aligned} &\frac{1}{\not{p} + \not{k} - m + i\epsilon} (p+m)(p-m+p+i\epsilon) = p^2 - m^2 + i\epsilon \\ &\downarrow \\ &(\not{p} + m) \gamma^\nu u(P, s) = (2p^\nu - \gamma^\nu \not{p} + \gamma^\nu m) u(P, s) = 2p^\nu u(P, s) \end{aligned}$

散射截面的量纲, 唯象解释?]

$$\begin{aligned} &\downarrow \\ &= \frac{q^4}{4m^2} \sum_{ss'} [\bar{u} Q_{\mu\nu} u]^\dagger [\bar{u} Q^{\mu\nu} u] \\ &= \frac{q^4}{4m^2} \text{tr}[u u^\dagger Q^{\mu\nu} u \bar{u} Q_{\mu\nu}] \\ &= \frac{q^4}{4m^2} \text{tr}[(\not{p}' + m) Q^{\mu\nu} (\not{p} + m) Q_{\mu\nu}] \\ &= \frac{q^4}{4m^2} \text{tr}[\not{p}' Q^{\mu\nu} \not{p} Q_{\mu\nu}] + m^2 [Q^{\mu\nu} Q_{\mu\nu}] \end{aligned}$$

$$\begin{aligned} O^{\mu\nu} u(P, s) &= \left[\gamma^\mu \frac{i(\not{p} \gamma^\nu + 2p^\nu)}{(\not{p} + \not{k})^2 - m^2 + i\epsilon} + \gamma^\nu \frac{i(2p^\mu - \not{k} \gamma^\mu)}{(\not{p} - \not{k}')^2 - m^2 + i\epsilon} \right] u(P, s) \\ &= \frac{i}{2m} \left[\frac{1}{\omega} (\gamma^\mu \not{k} \gamma^\nu + 2\gamma^\mu p^\nu) + \frac{1}{\omega'} (\gamma^\nu \not{k}' \gamma^\mu - 2\gamma^\nu p^\mu) \right] u(P, s) \\ &= \frac{i}{2m} \underbrace{Q^{\mu\nu}}_{\frac{\gamma^\mu \not{k} \gamma^\nu + 2\gamma^\mu p^\nu}{\omega} + \frac{\gamma^\nu \not{k}' \gamma^\mu - 2\gamma^\nu p^\mu}{\omega'}} u(P, s) \end{aligned}$$

求迹: 略

$$\sum |M_{fi}|^2 = 8e^4 \left[\frac{w'}{w} + \frac{w}{w'} + 2\left(\frac{m}{w} - \frac{m}{w'}\right) + \left(\frac{m}{w} - \frac{m}{w'}\right)^2 \right]$$

$$\downarrow = 8e^4 \left(\frac{w'}{w} + \frac{w}{w'} - \sin^2 \theta \right)$$

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2m^2} \frac{w'^2}{w^2} \left(\frac{w'}{w} + \frac{w}{w'} - \sin^2 \theta \right) \quad \alpha = \frac{e^2}{4\pi}$$

$$\begin{aligned} P' + k' &= P + k \\ P'^2 + k'^2 &= P^2 + k^2 \end{aligned} \rightarrow \begin{aligned} P'^2 + k'P' &= PP' + kP' \\ P'P + k'P &= P^2 + kP \\ P'k + k'k &= Pk + k^2 \\ P'k' + k'^2 &= Pk' + k'k' \end{aligned} \rightarrow \begin{aligned} PP' &= m^2 + kk' \\ Pk &= Pk' \\ P'k' &= Pk \end{aligned} \quad \begin{aligned} P &= (m, 0, 0, 0) \\ k &= (w, \mathbf{k}) \end{aligned}$$

$$\sum |M_{fi}|^2 = 8e^4 \left(\frac{Pk'}{Pk} + \frac{Pk}{Pk'} + \left(1 + \frac{P^2}{Pk} - \frac{P'^2}{Pk'} \right)^2 - 1 \right)$$

可由经典电动力学得到, 即:

$$\text{若不对光子求平均: } \frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4m^2} \frac{w'^2}{w^2} \left(4(e_{k'}^{\mu} e_{k\mu}^0)^2 + \frac{w'}{w} + \frac{w}{w'} - 2 \right)$$

经典电动力学为QED的2阶近似

$$\text{低能近似: } \frac{d\sigma}{d\Omega} = \frac{\alpha^2}{m^2} (e_{k'}^{\mu} e_{k\mu}^0)^2 \rightarrow \text{与偏振有关与波长无关}$$

$$= \frac{\alpha^2}{2m^2} (1 + \cos^2 \theta) \rightarrow \sigma_T = \int d\Omega \frac{\alpha^2}{2m^2} (1 + \cos^2 \theta) = \frac{8\pi}{3} r_e^2$$

$$\text{高能近似: } w \gg m, \quad P(E_f) = \frac{w' dw'}{\pi^2 dE_f} = \frac{w'^2}{\pi^2} \frac{P_0'}{P_0 + w}$$

高能: 螺旋性不变 $\rightarrow$ 散射角变大

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 w'}{4P_0(P_0 + w)w} \left[\frac{Pk'}{Pk} + \frac{Pk}{Pk'} + \left(\frac{P^2}{Pk} - \frac{P'^2}{Pk'} + 1 \right)^2 - 1 \right]$$

$$\text{质心系: } Pk = w(P_0 + w), \quad Pk' = w'(P_0 + w \cos \theta), \quad \frac{d\sigma}{d\Omega} \approx \frac{\alpha^2}{4P_0(P_0 + w \cos \theta)}, \quad \sigma_T \approx \frac{2\alpha^2}{2P_0 w} \ln \frac{P_0 + w}{P_0 - w}, \quad r_e = \frac{\alpha}{m} = \frac{e^2}{4\pi m}$$

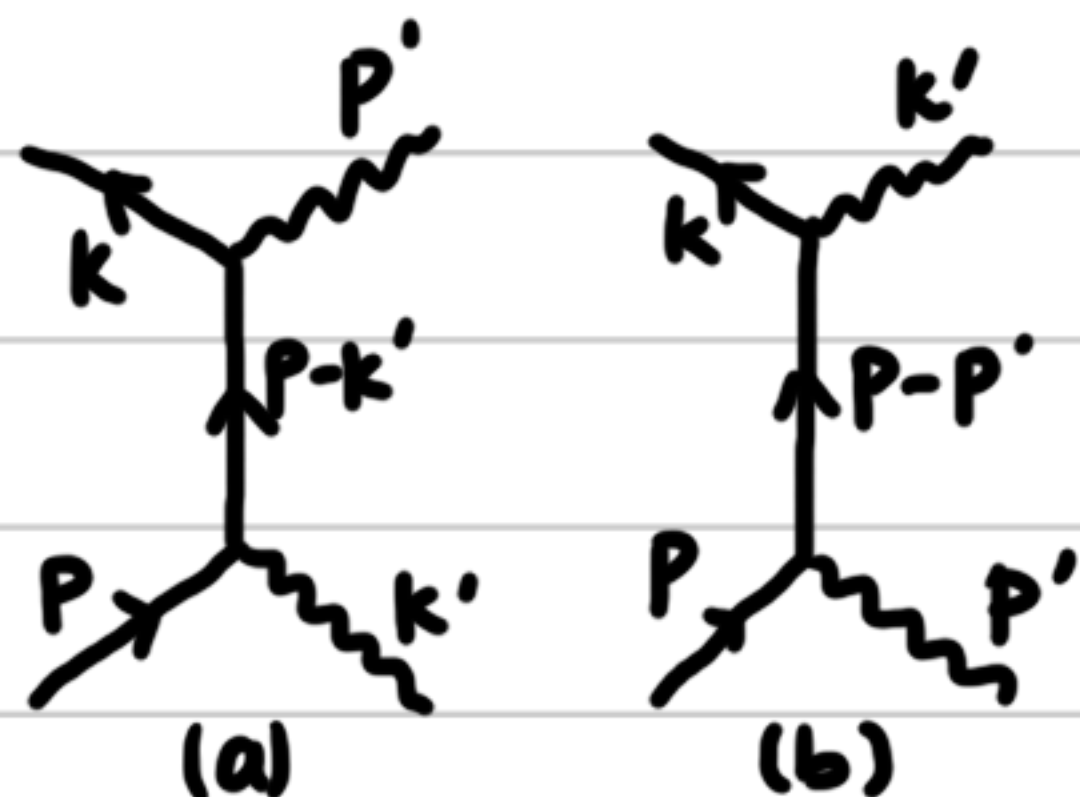
逆Compton 散射?

2. e^+e^- 湮灭

物理过程: $e^+e^- \rightarrow \gamma' + \gamma'' \dots$ (虚拟光子在质心系不满足动量守恒 ($\vec{p} = \vec{k} = 0$))

最小: 2阶: $e^+e^- \rightarrow \gamma' + \gamma''$ 以 e^- 为静止, $P=0, P_0=m$

动量 $k + P \rightarrow k' + P'$ 能量: $E_f = k_0' + P_0'$, $P(E_f) = \frac{k_0' dk_0'}{\pi^2 dE_f} = \frac{k_0'^3 P_0'}{\pi^2 m E_f}$, 归一化常数 $N_{fi} = \frac{1}{4\sqrt{P_0' k_0' m k_0}} V^2$



Feynmann 图: 2个, M_{fi} 有:

$$M_{fi(a)} = -ie^2 e_{P'\mu}^0 \bar{V}(k, s') \gamma^\mu \frac{i}{\not{P} - \not{k}' - m + i\epsilon} \gamma^\mu e_{k\mu}^0 U(P, s)$$

$$M_{fi(b)} = -ie^2 e_{k'\mu}^0 \bar{V}(k, s') \gamma^\mu \frac{i}{\not{P} - \not{P}' - m + i\epsilon} \gamma^\mu e_{P'\mu}^0 U(P, s)$$

$$M_{fi} = -ie^2 e_{P'\mu}^0 \bar{V}(k, s') O^{\mu\nu}(P-k', P-P') e_{k\mu}^0 U(P, s)$$

$$O^{\mu\nu}(P, q) = \gamma^\mu \frac{i}{\not{P} - m + i\epsilon} \gamma^\nu + \gamma^\nu \frac{i}{\not{q} - m + i\epsilon} \gamma^\mu$$

变量数的关系, 费曼图程序? □

同上, 对光子缩并:

$$\begin{aligned}\Sigma |M_{fi}|^2 &= e^4 \sum_{ss'} e_{p's'}^\mu \cdot e_{k's}^\mu e_{p'v}^\nu \cdot e_{k'v}^\nu \sum_{ss'} [\bar{v} O_{\nu\nu} u] [\bar{u} O^{\mu\mu} v] \\ &= e^4 \sum_{ss'} u O_{\mu\nu} \bar{v} \bar{u} O^{\mu\nu} v \\ &= e^4 \text{tr} [v(k,s') \bar{v}(k,s') O^{\mu\nu}(p-k', p-p') u(p,s) \bar{u}(p,s) O_{\mu\nu}(p-k', p-p')]\end{aligned}$$

与 Compton 散射类比:

代换: $k \rightarrow -p', p' \rightarrow -k, \bar{u}(p,s) \rightarrow \bar{v}(k,s)$

↓
 $\sum_s u(p,s) \bar{u}(p,s) = \not{p} + m \rightarrow \sum_s v(k,s) \bar{v}(k,s) = \not{k} - m$
↓

$$\Sigma |M_{fi}|^2 = 8e^2 \left[\frac{pk'}{pp'} + \frac{pp'}{pk'} - \left(\frac{p^2}{pp'} + \frac{p'^2}{pk'} - 1 \right)^2 + 1 \right]$$

$$\text{截面: } \frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2m^2} \frac{k_i^2}{|k|E_f} \left[\frac{pk'}{pp'} + \frac{pp'}{pk'} - \left(\frac{p^2}{pp'} + \frac{p'^2}{pk'} - 1 \right)^2 + 1 \right]$$

$$E_f = k'_0 + p'_0$$

$$= \frac{r_e^2}{2m} \frac{k_i^4}{|k|E_f} [E_f(1 - \cos\Theta) - m(1 + \cos^2\Theta)] \quad ? \square \text{ 何来 } \cos\Theta$$

交叉对称性

compton: $\gamma + e \rightarrow \gamma' + e'$

对 S 矩阵: Bose 子产生湮灭的外线贡献相同

湮灭: $e + e' \rightarrow \gamma + \gamma'$

Fermi 子产生/湮灭 $\bar{u} \rightarrow \bar{v}$

↓
把一端粒子变成另一端反粒子

3. $e^+e^- \rightarrow \mu^+\mu^-$ 过程

物理过程: $e^+ + e^- \rightarrow \mu^+ + \mu^-$

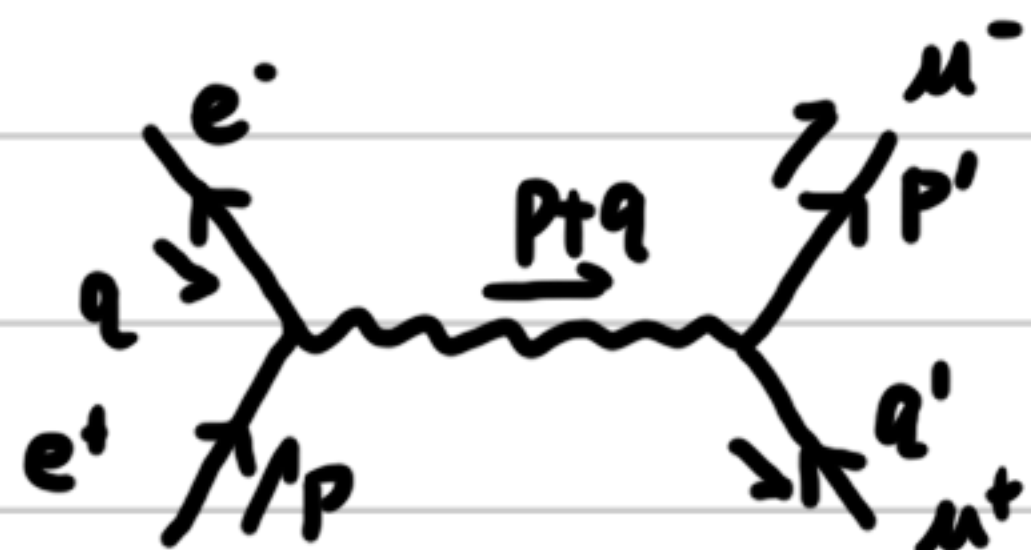
拉氏密度: $\mathcal{L}_I = e \bar{\psi}_e \gamma^\mu \psi_e A_\mu + e \bar{\psi}_\mu \gamma^\mu \psi_\mu A_\mu$

$$M_{fi} = -ie^2 \bar{u}(p',s') \gamma^\nu v(q',s') iD_F(p+q) \bar{v}(q,s) \gamma^\mu u(p,s) \quad (\text{带'的为 } \mu, \text{不带为 } e)$$

$$= \frac{-e^2}{(p+q)^2} \bar{u}' \gamma^\mu v' \bar{v} \gamma_\mu u$$

$$\begin{aligned}|M_{fi}|^2 &= \frac{e^4}{(p+q)^4} \text{tr} [\bar{u} \gamma_\mu v \bar{v}' \gamma^\mu u' \bar{u}' \gamma^\nu v' \bar{v} \gamma_\nu u] = \frac{e^4}{s^2} \text{tr} [\bar{u} \gamma_\mu v \cdot \bar{v}' \gamma_\nu u'] \text{tr} [\bar{v}' \gamma^\mu u' \bar{u}' \gamma^\nu v'] \\ &= \frac{e^4}{s^2} \text{tr} (u \bar{u} \gamma_\mu v \bar{v}') \text{tr} (\bar{v}' v' \gamma^\mu \bar{u}' u' \gamma^\nu)\end{aligned}$$

二阶 Feynmann 图:



$$\begin{aligned}
\frac{1}{4} \sum_{ss's's'} |M_{fi}|^2 &= \frac{e^2}{4s^2} \text{tr}((q-m)\gamma_\nu(p+m)\gamma_\mu) \text{tr}((q'-m)\gamma^\mu(p'+m)\gamma^\nu) \\
&= \frac{e^2}{4s^2} (\text{tr}(\gamma_\nu p \gamma_\mu) - m^2 \text{tr}(\gamma_\mu \gamma_\nu) + m \text{tr}(\underbrace{q \gamma_\nu \gamma_\mu - \gamma_\nu q \gamma_\mu}_0) \cdot (\text{tr}(q' \gamma^\mu p' \gamma^\nu) - m^2 \text{tr}(\gamma^\mu \gamma^\nu) + m \text{tr}(\underbrace{q' \gamma^\mu \gamma^\nu - \gamma^\mu p' \gamma^\nu}_0)) \\
&= \frac{e^2}{4s^2} (4(q_\nu p_\mu + p_\nu q_\mu - (m^2 + q p) g_{\mu\nu})) (4(q'^\mu p'^\nu + p'^\mu q'^\nu - (m'^2 + q' p') g^{\mu\nu})) \\
&= \frac{2e^2}{s^2} (p q' \cdot q p' + p p' \cdot q q' + p q m'^2 + p' q' m^2 + 2 m m')
\end{aligned}$$

Mandelstam 变量: $s = (p+q)^2 = \sum_i m_i^2 = \sum E_i^2$ (Lorentz 不变的) $t = (p-p')^2$, $u = (p-q')^2$

$\downarrow \hookrightarrow s+t+u = p^2 + q^2 + p'^2 + q'^2 = \sum m_i^2 \rightarrow 2$ 个独立

$p q = \frac{s}{2} - m^2$, $p' q' = \frac{s}{2} - m'^2$; $p p' = q q' = \frac{1}{2}(t - m^2 - m'^2)$

$p q' = q p' = \frac{1}{2}(u - m^2 - m'^2) = \frac{1}{2}(s+t - m^2 - m'^2)$

$$\frac{1}{4} \sum |M_{fi}|^2 = \frac{2e^4}{s^2} (2s(m^2 + m'^2) + (t - m^2 - m'^2)^2 + (s+t - m^2 - m'^2)^2)$$

质心系不变微分截面 $\vec{p} + \vec{q} = \vec{p}' + \vec{q}' = 0 \rightarrow q_0 = p_0 = q'_0 + p'_0 = \frac{E_f}{2}$, $v = \frac{2|\vec{p}|}{p_0} = \frac{4|\vec{p}|}{E_f}$

有: $N_{fi} = \frac{1}{4|\vec{p}_0 q_0 p'_0 q'_0| v} = \frac{1}{E_f^4 v}$, $\frac{dE_f}{d|\vec{p}|} = 2 \frac{d}{d|\vec{p}|} \sqrt{|\vec{p}|^2 + m^2} = \frac{4|\vec{p}|}{E_f}$

Lorentz 不变

$\rho(E_f) = \frac{p^2 dp}{\pi^2 dE_f} = \frac{|\vec{p}| E_f}{4\pi^2}$

$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 |\vec{p}|^2}{2s^2 |\vec{p}|} [2s(m^2 + m'^2) + (t - m^2 - m'^2)^2 + (s+t - m^2 - m'^2)^2]$

$\downarrow$ 与 4π 乘 $d\Omega = \sin\theta d\theta d\varphi$

$\frac{d\sigma}{\sin\theta d\theta} = \frac{2\pi d\sigma}{d\Omega}$, $\vec{p}^2 = p_0^2 - m^2 = \frac{s-4m^2}{4}$, $dt = d(p-p')^2 = d(m^2 + m'^2 + 2p_0 p'_0 - 2|\vec{p}||\vec{p}'|\cos\theta) = 2|\vec{p}||\vec{p}'|\sin\theta d\theta$

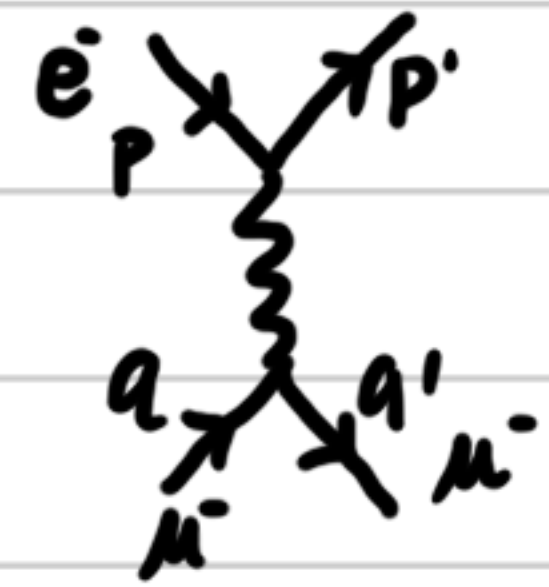
$\frac{d\sigma}{dt} = \frac{2\pi\alpha^2}{s^2(s-4m^2)} [2s(m^2 + m'^2) + (t - m^2 - m'^2)^2 + (s+t - m^2 - m'^2)^2]$

高能: $s, |t|_{\max} \gg m^2, m'^2$

$$\downarrow |t| \gg m^2 \rightarrow \frac{d\sigma}{dt} \approx \frac{2\pi\alpha^2}{s^4} (s^2 + 2t^2 + 2st) \quad , \quad t \text{ 积分有: } t(\theta=0) = t_b \approx 0 \quad , \quad t(\theta=\pi) = t_a \approx -s$$

$$\sigma = \int_{t_a}^{t_b} \frac{d\sigma}{dt} \approx \frac{2\pi\alpha^2}{s^4} \cdot \frac{2}{3}s^3$$

4. eμ散射与Mott散射



eμ散射: $\bar{e}\mu \rightarrow e\bar{\mu}$: $\left. \begin{array}{l} \bar{e}\mu \rightarrow e\bar{\mu} \\ e^+e^- \rightarrow \mu^+\mu^- \end{array} \right\} \rightarrow \text{交叉对称性: } p \rightarrow p, q \rightarrow -p', p' \rightarrow q', q' \rightarrow -q$

$$4\pi|M_{fi}|^2 = \frac{8e^4}{t^2} [(pq)^2 + (p'q')^2 - (pp' - m^2)(m^2 + m'^2)]$$

能动: $p'q' = pq, pq' = p'a, qq' = pp' + m'^2 - m^2, E_f = p'_0 + q'_0 = \sqrt{\vec{p}'^2 + m^2} + \sqrt{\vec{q}'^2 + m'^2}$

取μ静止: $q = (m', 0, 0, 0)$, 有: $\frac{dE_f}{dp'_1} = \frac{E_f(p'_1 - p_0|p|\cos\theta)}{p'_0 q'_0}, \quad P(E_f) = \frac{|\vec{p}'|^2 dp'_1}{\pi^2 dE_f} = \frac{\vec{p}'^2 p'_0 q'_0}{\pi^2 (E_f |p'_1 - p_0|p|\cos\theta)}$

(是否可用图退化来解释 QED $\rightarrow$ ED 的过程?)

$$\downarrow \frac{d\sigma}{d\Omega} = \frac{V^4}{16v^2} P(E_f) N_f^2 \sum |M_{fi}|^2 = \frac{2\alpha^4 |p|}{m' (E_f |p'_1 - p_0|p|\cos\theta) t^2} [(p_0^2 + p_0'^2) m'^2 - (pp' - m^2)(m^2 + m'^2)]$$

Mott散射: $m' \rightarrow \infty$, 电子被点电荷Columb场散射 $|p'| = |p|, E_f \approx m'$

$$\frac{d\sigma}{d\Omega} \stackrel{\downarrow}{=} \frac{2\alpha^4}{t^2} (p_0^2 + p_0'^2 - pp' + m^2) = \frac{2\alpha^4}{16p^4 \sin^4(\frac{\theta}{2})} 2p_0^2 (1 - v^2 \sin^2(\frac{\theta}{2}))$$

$$\downarrow v \ll 1 \text{ 为Rutherford散射} = \frac{\alpha^2}{4p^2 v^2 \sin^4(\frac{\theta}{2})} (1 - v^2 \sin^2(\frac{\theta}{2}))$$

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4m^2 v^2 \sin^4(\frac{\theta}{2})}$$

4. 等效外场近似

1 等效外场: Coulomb 场: 用 Mott 散射法, $\mu \rightarrow m'$, Ze 粒子, 供能 $-iZe(2\pi)^4 \delta^4(q'-q-k) \bar{u}(q', s') \gamma^\mu u(q, s) iD_{\mu\nu}(k)$

$$j^\mu = Ze \bar{\psi}(x) \gamma^\mu \psi(x) \Rightarrow m' \rightarrow \infty \text{ 时 } u(q, s) = N \begin{pmatrix} \xi_s \\ \frac{\mathbf{q} \cdot \boldsymbol{\sigma}}{m'+m} \xi_s \end{pmatrix} \rightarrow N \begin{pmatrix} \xi_s \\ 0 \end{pmatrix}$$

$$\bar{u}(q', s') \gamma^0 u(q, s) = N^2 (\xi_{s'}^\dagger \cdot 0) (1 \cdot -1) (\xi_s) = N^2 \xi_{s'}^\dagger \xi_s = \delta_{s's}$$

$$\bar{u}(q', s') \gamma^i u(q, s) = 0$$

Feynmann 图的受力机理? $\square$

自旋耦合 $\rightarrow s=s'$ 忽略

$$\rightarrow \text{原式为: } -iZe(2\pi)^4 \delta^4(q'-q-k) \delta_{s's} iD_{F\mu 0}(k)$$

$$\hookrightarrow \text{对 } k_0 \text{ 积分: } -iZe(2\pi)^3 \delta^3(\mathbf{q}'-\mathbf{q}-\mathbf{k}) iD_{F\mu 0}(k)$$

$$\text{对 } \mathbf{q} \text{ 平均, 对 } \mathbf{q}' \text{ 求和: } \int d\mathbf{q} \rightarrow iZe \cdot iD_{F\mu 0}(k) = \frac{Ze}{k^2} g_{\mu 0} = \frac{Ze}{k^2} g_{\mu 0}$$

$$\text{等效外场 } A_0^{\text{ext}}(x) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{Ze}{k^2} e^{-ikx} = \frac{Ze}{4\pi r} \rightarrow \text{静电势}$$

Coulomb 场: 外源 $A_\mu^{\text{ext}}(k) = \frac{Ze}{k^2} g_{\mu 0} \rightarrow \text{Feynmann } \square \times$

(加一个对 k 求和)

外源引起 Coulomb 散射: (存在一阶 S 图)

$$S_{fi}'' = -ie N_{fi} \int \frac{d^4k}{(2\pi)^4} (2\pi)^4 \delta^4(p'-p-k) \delta^3(p'-p-k) \bar{u}(p', s') \gamma^\mu u(p, s) \frac{Ze}{k^2} g_{\mu 0} \\ = -i2\pi \delta(p'-p) \frac{Ze^2}{(p'-p)^2} N_{fi} u^\dagger(p', s') u(p, s) = -i2\pi \delta(p'-p) N_{fi} M_{fi} \Rightarrow M_{fi} = \frac{Ze^2}{(p'-p)^2} u^\dagger(p', s') u(p, s)$$

$$\text{能量守恒 (此处无动量)} \rightarrow |\mathbf{p}'| = |\mathbf{p}| \Rightarrow (\mathbf{p}' - \mathbf{p})^2 = 4p^2 \sin^2 \frac{\theta}{2}$$

$$\text{跃迁率: } W_{fi} = |S_{fi}''|^2 = (2\pi)^4 \delta(0) \delta(p'-p) N_{fi}^2 |M_{fi}|^2 \cdot g_{00} = \lim_{T \rightarrow \infty} \int \frac{dt}{2\pi} e^{i p_0 t} = \lim_{T \rightarrow \infty} \frac{T}{2\pi}$$

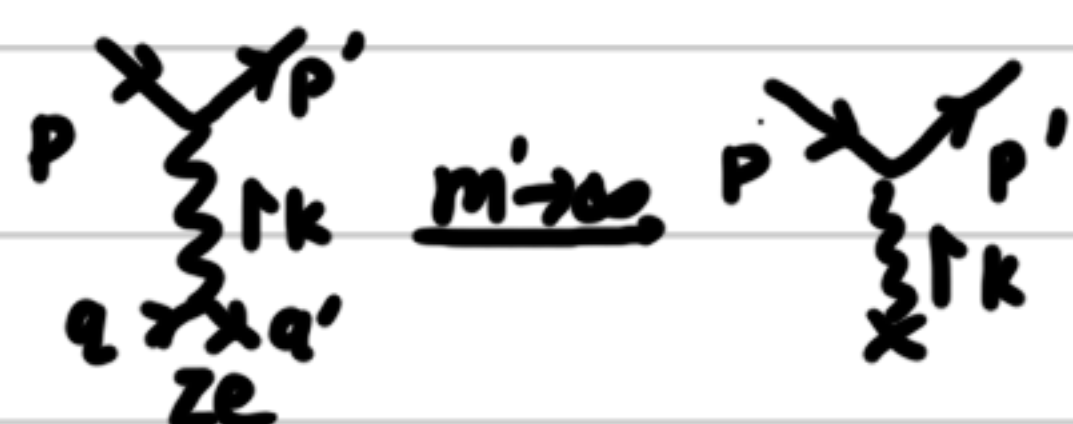
$$P_{fi} = \lim_{T \rightarrow \infty} \frac{W_{fi}}{T} = 2\pi \delta(p'-p) N_{fi}^2 |M_{fi}|^2 \quad \text{此处书中少一个 } 2\pi \text{ 于 } \delta(p'-p)$$

$$\bar{P}_{fi} = \frac{1}{2} \sum_{s, s'} \int \frac{V d^3p}{(2\pi)^3} 2\pi \delta(p'-p) N_{fi}^2 |M_{fi}|^2 = \frac{V}{2\pi^2} \int d\Omega |\mathbf{p}| p_0 N_{fi}^2 |M_{fi}|^2$$

$$\sum_s u(p, s) \bar{u}(p, s) = \not{p} + m \\ \sum_s u u^\dagger = \sum u \bar{u} \gamma^0 = (\not{p} + m) \gamma^0$$

$$\sum u^\dagger(p, s) u(p', s') u^\dagger(p', s') u(p, s) = \sum_s \bar{u}(p, s) \gamma^0 (\not{p}' + m) \gamma^0 u(p, s)$$

$$\sum |M_{fi}|^2 = \frac{Ze^4}{2p^2 v \sin^4(\frac{\theta}{2}) (1 - v^2 \sin^2 \frac{\theta}{2})} = \text{tr}((\not{p} + m) \gamma^0 (\not{p}' + m) \gamma^0) = 4(2p_0 p'_0 - \mathbf{p} \cdot \mathbf{p}' + m^2)$$



电磁场 4 维势是否可作为 A_μ^{ext} ? $\square$



$$\frac{d\sigma}{d\Omega} = \frac{V}{v} \frac{d\bar{R}_i}{d\Omega} = \frac{v^2}{4\pi v} \sum N_{fi}^2 |N_{fi}|^2 = \frac{Z^2 \alpha^2}{4\pi v^2 \sin^2(\frac{\theta}{2})} [1 - v^2 \sin^2(\frac{\theta}{2})] \rightarrow \text{结果同Mott散射}$$

$N_{fi} = \frac{1}{2\sqrt{p_i p_f} V}$

轫致辐射



$$e^- + N \rightarrow e^- + N + \gamma$$

2种2阶S矩阵: $S_{fi}^{(2)} = ie N_{fi} \int \frac{d^4 q}{(2\pi)^4} \frac{d^3 \vec{k}}{(2\pi)^3} (2\pi)^4 \delta^4(P+k-q) \bar{u}(p', s') \gamma^\mu e_{k, \mu}^\dagger S_F(q) \cdot ie (2\pi)^4 \delta^4(q-P-k) \delta^3(\vec{q}-\vec{P}-\vec{k}) \gamma^\mu u(p, s) \frac{Ze}{k^2} g_{\nu 0}$

$= -i 2\pi \delta(P+k_0-P_0) N_{fi} M_{fi(a)}$

$$M_{fi(a)} = \bar{u}(p', s') \gamma^\mu e_{k, \mu}^\dagger S_F(P+k) \gamma^0 u(p, s) \frac{Ze^3}{(\vec{P}+\vec{k}-\vec{P})^2}$$

$$= \frac{Ze^3 \bar{u}(p', s') \gamma^\mu e_{k, \mu}^\dagger (\not{P}+\not{k}+m) \gamma^0 u(p, s)}{[(P+k)^2 - m^2][(\vec{P}+\vec{k}-\vec{P})^2]}$$

类似

$$M_{fi(b)} = \frac{Ze^3 \bar{u}(p', s') \gamma^0 (\not{P}-\not{k}+m) \gamma^\mu e_{k, \mu}^\dagger u(p, s)}{[(P-k)^2 - m^2][(\vec{P}+\vec{k}-\vec{P})^2]}$$

$$M_{fi} = M_{fi(a)} + M_{fi(b)} = \frac{Ze^3 \bar{u}(p', s') Q u(p, s)}{(\vec{P}+\vec{k}-\vec{P})^2}, \quad Q = \frac{\not{P}+\not{k}+m}{2P_{k'}} \gamma^0 - \frac{\gamma^0 (\not{P}-\not{k}+m)}{2P_{k'}} \cdot Q = \gamma^\mu e_{k, \mu}^\dagger$$

$$\frac{1}{2} \sum_{s, s'} |M_{fi}|^2 = \frac{Z^2 e^6}{2(\vec{P}+\vec{k}-\vec{P})^4} \sum_{s, s'} (\bar{u}(p, s) Q u(p', s') \bar{u}(p', s') Q u(p, s))$$

$(\not{P}-m)u(p, s)=0 \rightarrow \bar{u} Q u = \bar{u}(p, s) Q u(p, s)$
 $\bar{u}(p', s')(\not{P}'-m)=0$
 $\gamma^\mu \not{p} = \not{p} \gamma^\mu$

$Q = \frac{(2E\not{P} + \not{k})\gamma^0}{2P_{k'}} - \frac{\gamma^0(2E\not{P} - \not{k})}{2P_{k'}} = \text{tr}[(\not{P}+m)Q(\not{P}+m)Q]$

Bethe-Heitler公式
 Kohn-Nishina公式

5. 标量场 QED

1. 相互作用: 定域规范不变性 $\rightarrow \mathcal{L} = D_\mu \phi^\dagger D^\mu \phi - m^2 \phi^\dagger \phi + \mathcal{L}_I$

$$\mathcal{L}_S = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi$$

$$\mathcal{L}_I = \underbrace{-iq\phi^\dagger \overleftrightarrow{\partial}_\mu \phi A^\mu}_{\text{导数耦合}} + \underbrace{q^2 \phi^\dagger \phi A^\mu A_\mu}_{\text{质量项}}$$

S矩阵: 对标量场: 同前, $i\Delta_F = \int \frac{d^4k}{(2\pi)^4} \frac{1}{p^2 - m^2 + i\epsilon} e^{-ikx}$

外源 $J^\dagger, J, J^\mu \rightarrow \mathcal{L}_I = q \frac{\partial}{\partial J^\dagger} \overleftrightarrow{\partial}_\mu \frac{\partial}{\partial J} + iq^2 g^{\mu\nu} \frac{\delta}{\delta J} \frac{\delta}{\delta J^\dagger} \frac{\delta}{\delta J^\mu} \frac{\delta}{\delta J^\nu}$

$$S = \mathcal{N} [e^{i\mathcal{L}_I} Z[\phi_{in}, \phi_{in}^\dagger, A_{\mu in}; J^\dagger, J, J^\mu]]$$

微扰展开

$$S_{(a)} = \mathcal{N} [e^{i\mathcal{L}_I} Z] \rightarrow S_{1a} = iq \mathcal{N} \int d^4x [\phi_{in}^\dagger(x) i \overleftrightarrow{\partial}^\mu \phi_{in}(x)] A_{\mu in}(x)$$

$$S_{(b)} = \mathcal{N} [e^{i\mathcal{L}_I} Z] \cdots S_{1b} = iq^2 \mathcal{N} \int d^4x \phi_{in}^\dagger(x) \phi_{in}(x) A_{\mu in}(x) A^\mu_{in}(x)$$

π - γ 散射:

$$\gamma + \pi \rightarrow \gamma' + \pi'$$

$$\text{能量守恒: } p+k = p'+k'$$

二阶 S 矩阵:

$$M_{fi}^{(2)a} = -iq^2 \epsilon'(p+k+p') i\Delta_F(p+k) (2p+k) \epsilon \quad (\epsilon = [\epsilon_{k\mu}^0], \epsilon' = [\epsilon_{k'\mu}^0])$$

$$M_{fi}^{(2)b} = -iq^2 \epsilon(p-k+p') i\Delta_F(p-k) (2p-k) \epsilon'$$

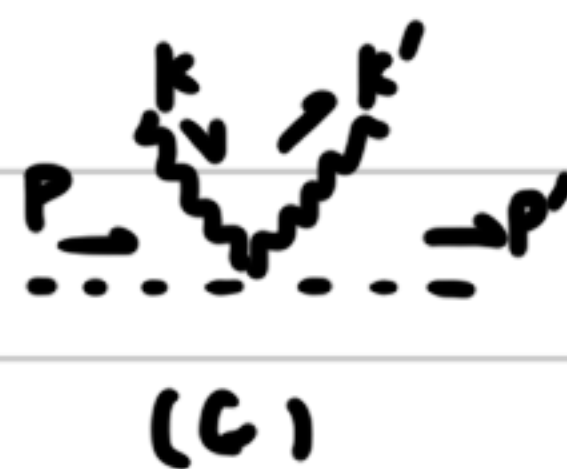
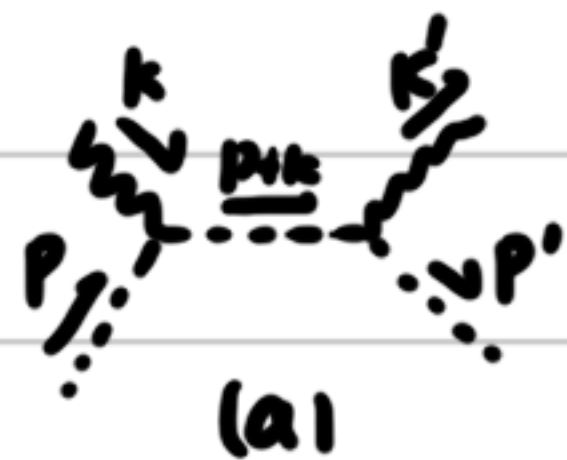
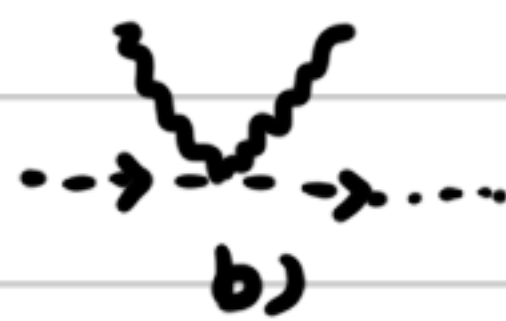
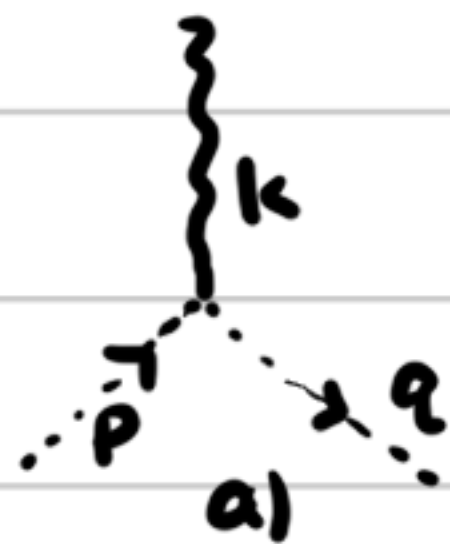
$$M_{fi}^{(2)c} = -2q^2 \epsilon_{k'\mu}^0 g^{\mu\nu} \epsilon_{k\mu}^0 = -2q^2 \epsilon' \epsilon$$

实验室系: $p = (m, 0, 0, 0)$, 横光子规范: $\vec{k} \cdot \vec{e}_{k0} = 0 \rightarrow$

$$\epsilon(2p+k) = -\vec{e}(2\vec{p}+\vec{k}) = -\vec{e} \cdot \vec{k} = 0$$

$$\epsilon(2p-k) = \vec{e} \cdot \vec{k} = 0$$

$$M_{fi} = M_{fi(c)} = -2q^2 \epsilon' \epsilon = 2q^2 (\vec{e}' \cdot \vec{e})$$



截面公式:

$$\left. \begin{aligned} \frac{d\sigma}{d\Omega} &= \frac{1}{(16\pi)^2} \frac{\omega'^2}{m^2 \omega^2} 4 |M_{fi}|^2 = \frac{\alpha^2 \omega'^2}{m^2 \omega^2} (\vec{e}' \cdot \vec{e})^2 \\ \text{Compton: } \frac{\omega'}{\omega} &= \frac{1}{[1 + \frac{\omega}{m}(1 - \cos\theta)]^2} \end{aligned} \right\} \frac{d\sigma}{d\Omega} = \frac{\alpha^2}{m^2} \frac{1 + \cos^2\theta}{2 [1 + \frac{\omega}{m}(1 - \cos\theta)]^2}$$

e平均, e'求和: $(\vec{e}' \cdot \vec{e})^2 \rightarrow 1 + \cos^2\theta$